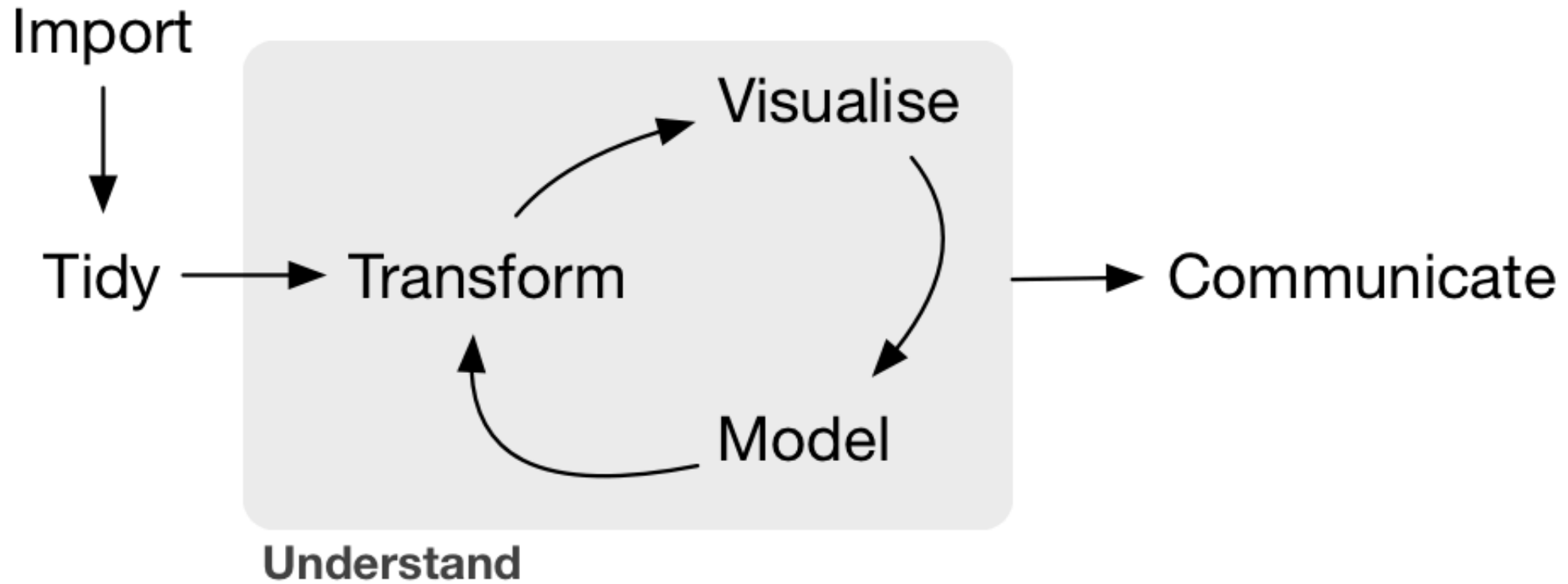


Inference for Proportions

IS381 - Statistics & Probability for Data Analytics

Jason Bryer, Ph.D.

Workflow



Source: Wickham & Grolemund, 2017

Tidy Data

“**TIDY DATA** is a standard way of mapping the meaning of a dataset to its structure.”

—HADLEY WICKHAM

In tidy data:

- each variable forms a column
- each observation forms a row
- each cell is a single measurement

each column a variable

id	name	color
1	floof	gray
2	max	black
3	cat	orange
4	donut	gray
5	merlin	black
6	panda	calico

each row
an
observation

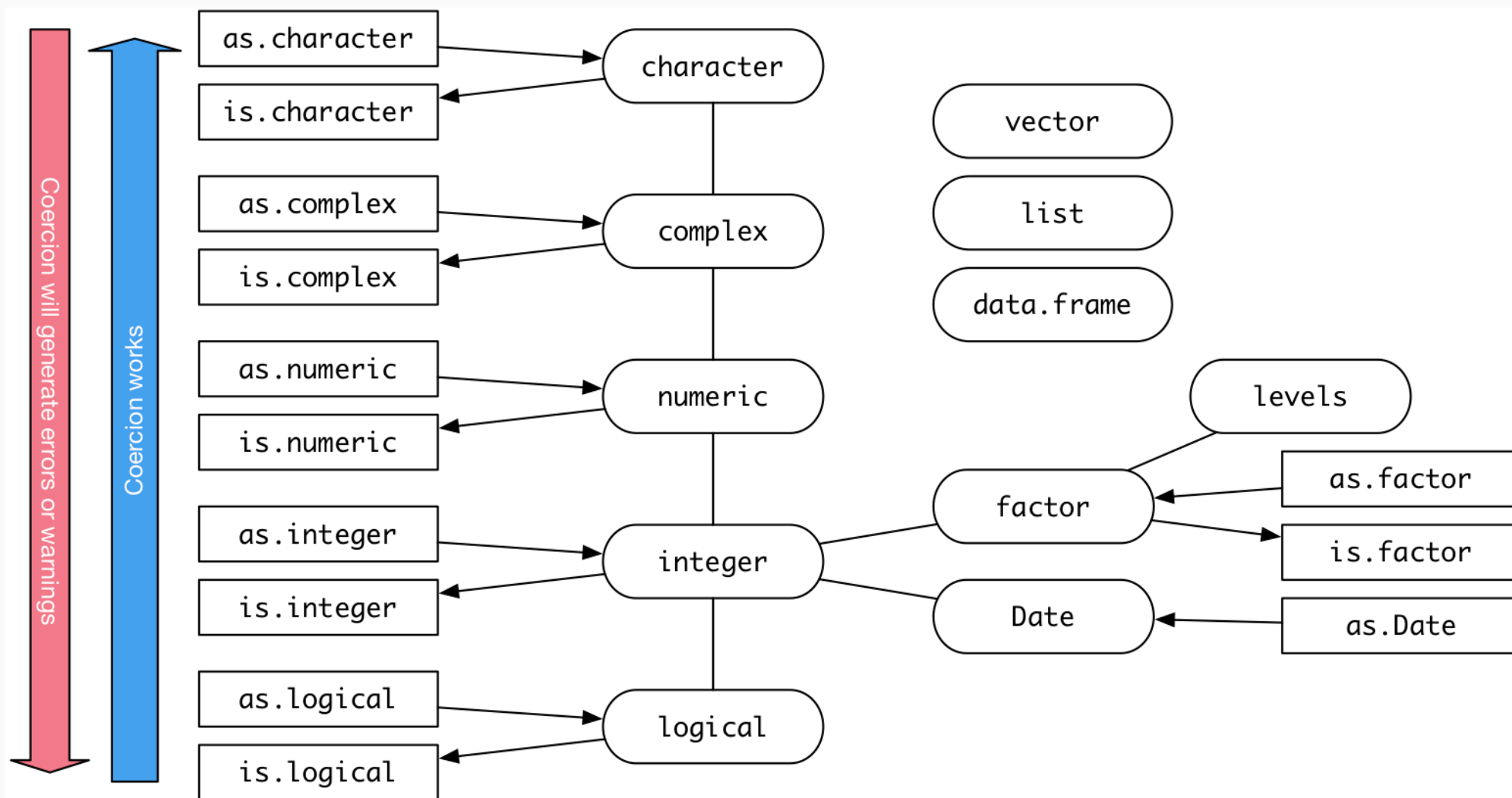
Wickham, H. (2014). Tidy Data. Journal of Statistical Software 59 (10). DOI: 10.18637/jss.v059.i10

Types of Data

- Numerical (quantitative)
 - Continuous
 - Discrete
- Categorical (qualitative)
 - Regular categorical
 - Ordinal



Data Types in R



Data Types / Descriptives / Visualizations

Data Type	Descriptive Stats	Visualization
Continuous	mean, median, mode, standard deviation, IQR	histogram, density, box plot
Discrete	contingency table, proportional table, median	bar plot
Categorical	contingency table, proportional table	bar plot
Ordinal	contingency table, proportional table, median	bar plot
Two quantitative	correlation	scatter plot
Two qualitative	contingency table, chi-squared	mosaic plot, bar plot
Quantitative & Qualitative	grouped summaries, ANOVA, t-test	box plot

Statistics

When describing a quantitative variable we are often interested in two things:

1. A measure of center
2. A measure of spread

The most common measures we will use in this class is the mean and median.

$$\bar{x} = \frac{\Sigma(x_i)}{n}$$

$$S^2 = \frac{\Sigma(x_i - \bar{x})^2}{N}$$

Note that in the numerator for the variance calculation we square the differences (also known as deviations). Squaring terms is common practice in statistics that serves two purposes:

1. It makes all the values positive.
2. It weighs observations that are further from the center more.

Variance

Population Variance:

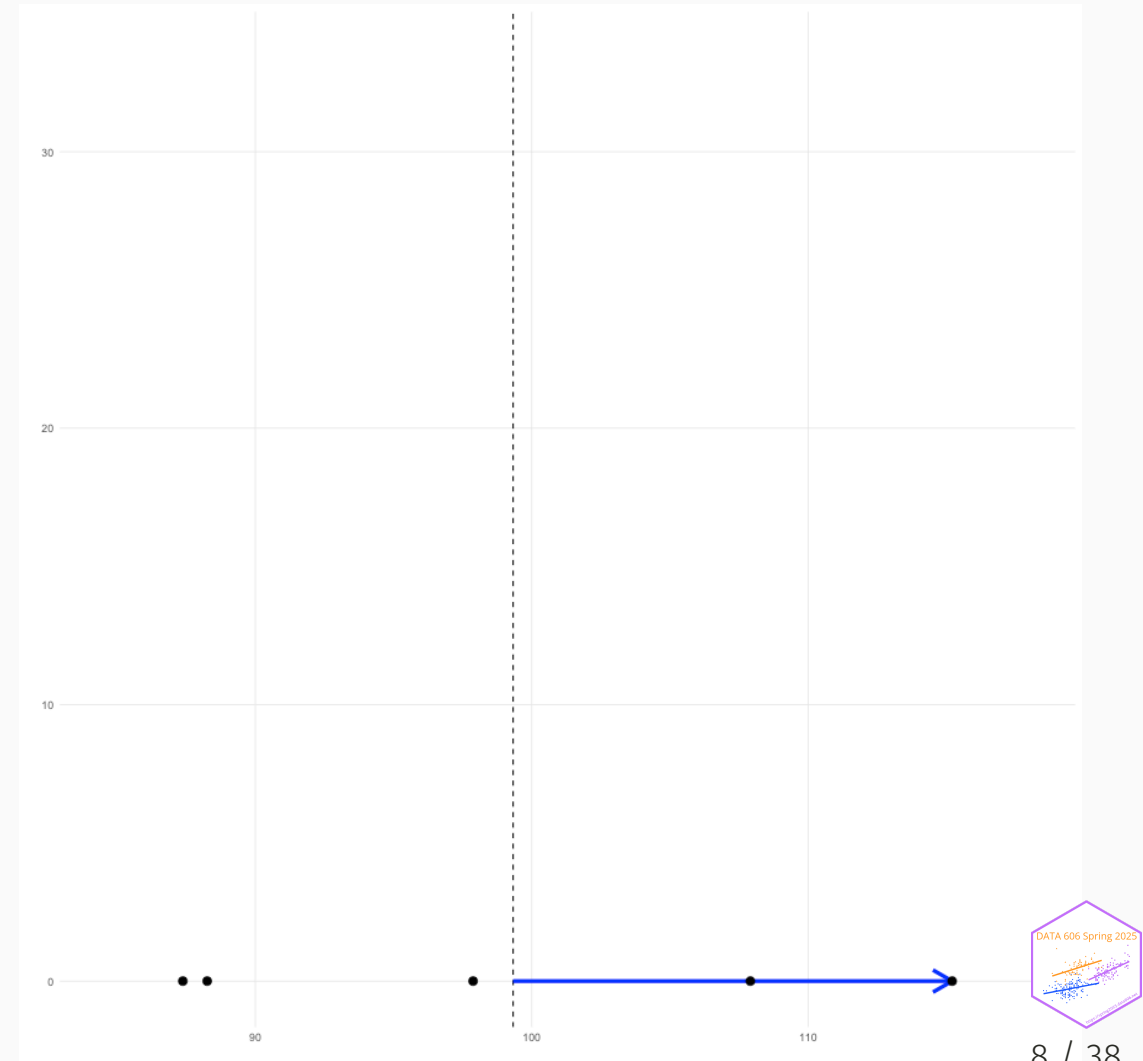
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Consider a dataset with five values (black points in the figure). For the largest value, the deviance is represented by the blue line ($x_i - \bar{x}$).

See also:

<https://shiny.rit.albany.edu/stat/visualizess/>

<https://github.com/jbryer/VisualStats/>

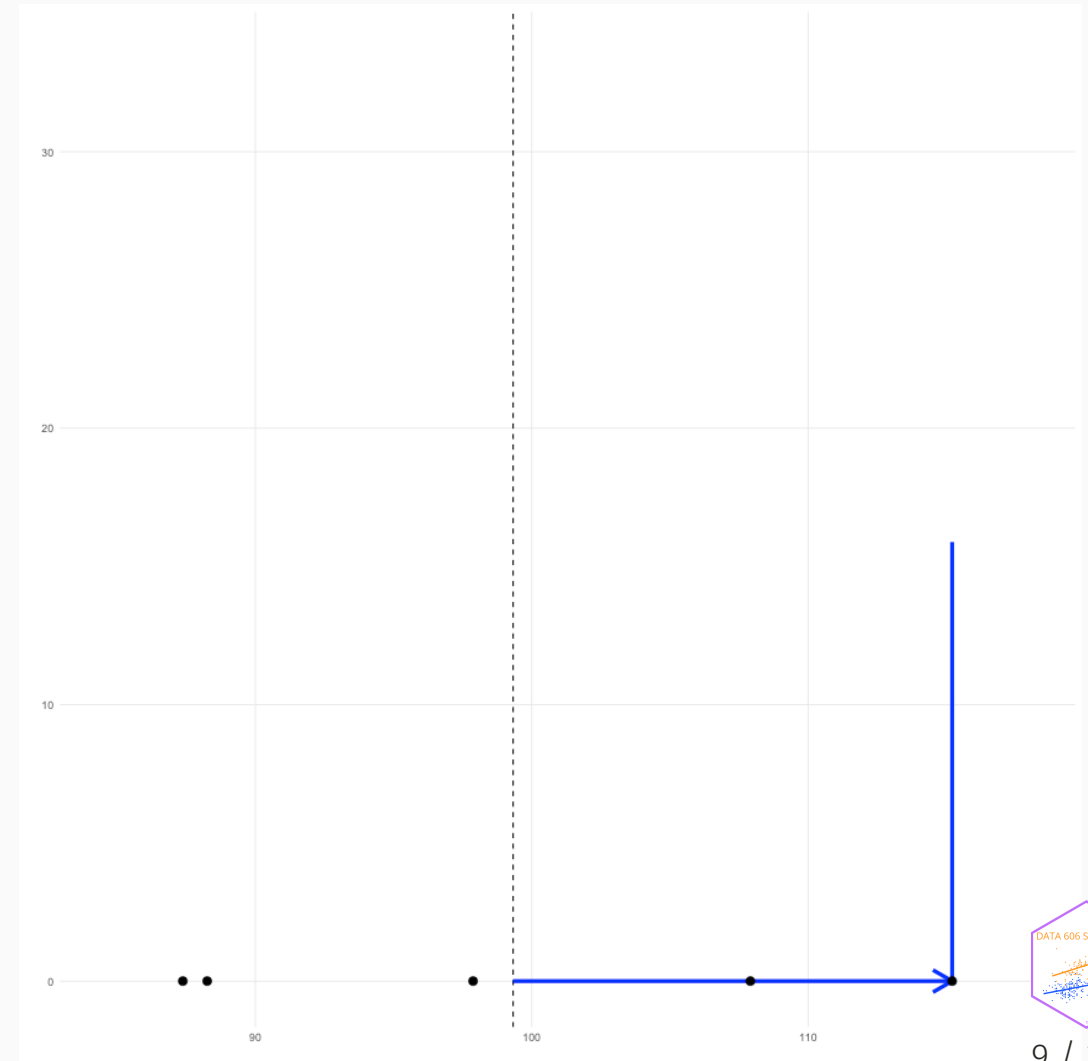


Variance (cont.)

Population Variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

In the numerator, we square each of these deviances. We can conceptualize this as a square. Here, we add the deviance in the y direction.

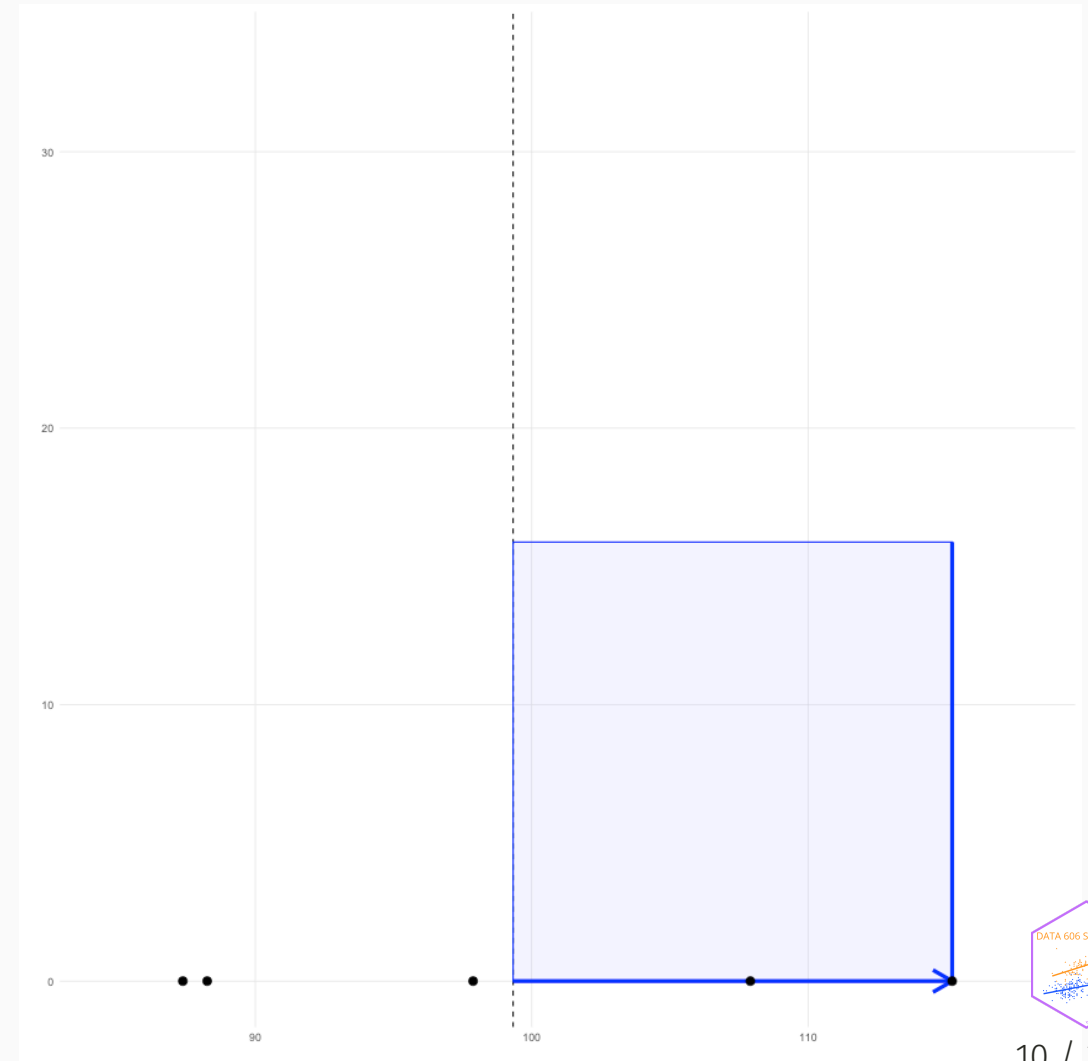


Variance (cont.)

Population Variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

We end up with a square.

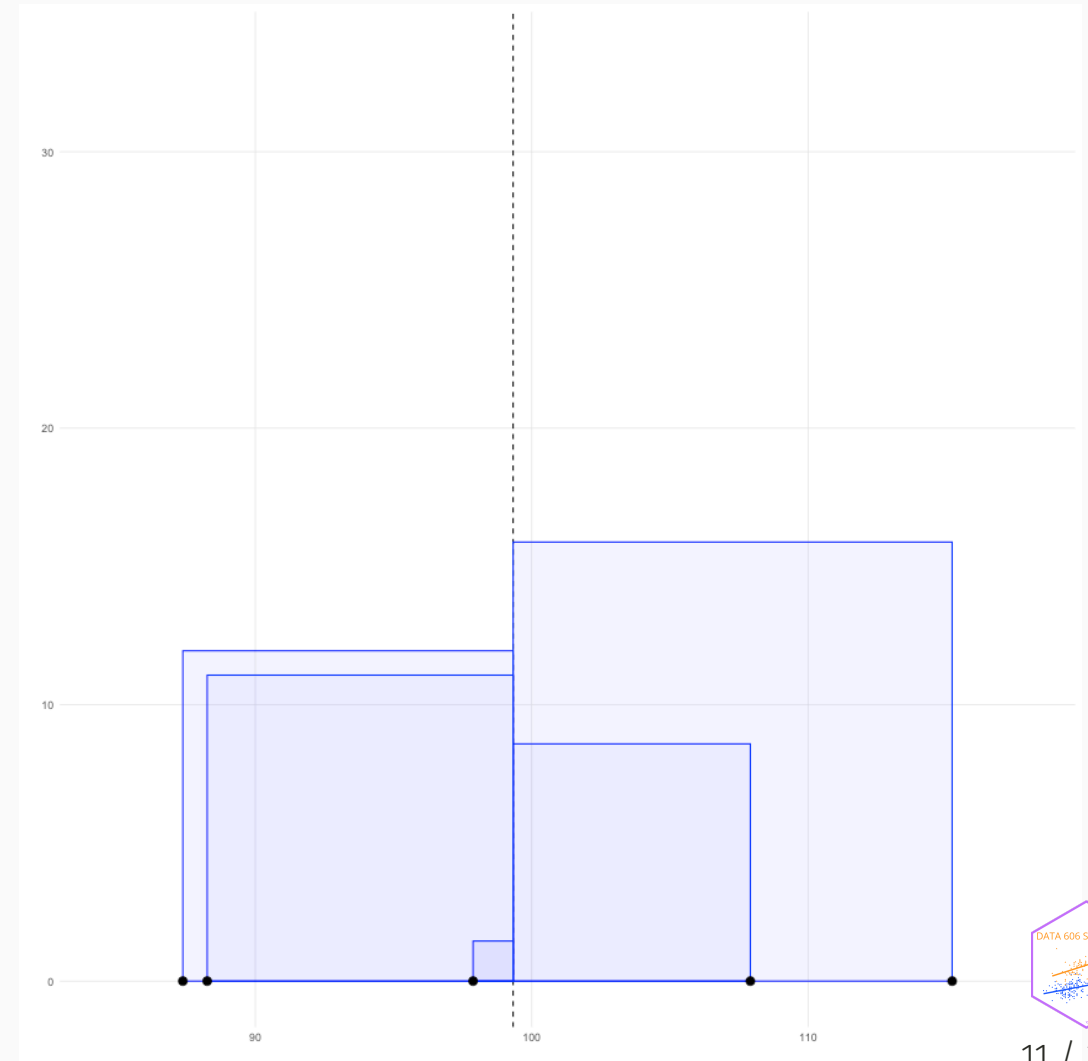


Variance (cont.)

Population Variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

We can plot the squared deviance for all the data points. That is, each component in the numerator is the area of each of these squares.

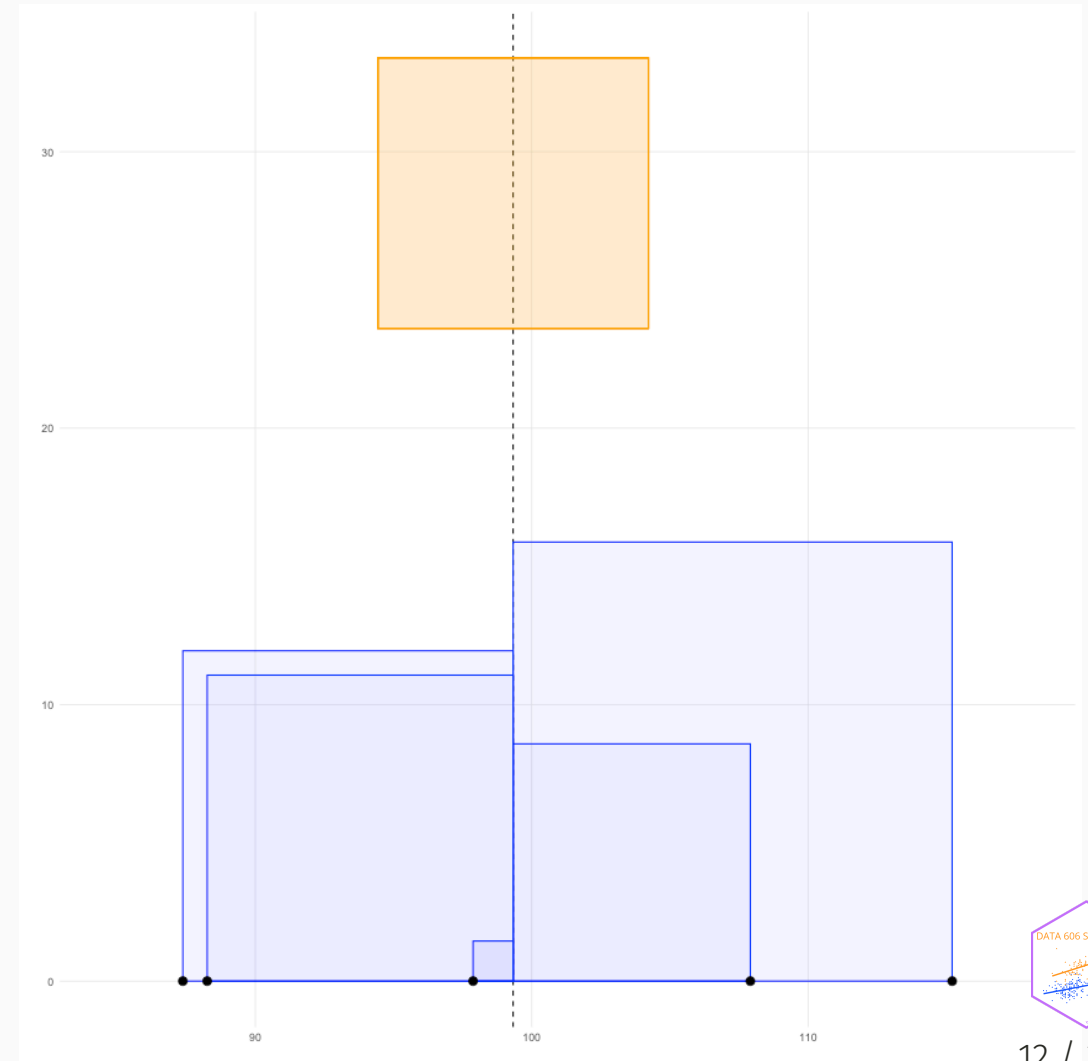


Variance (cont.)

Population Variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

The variance is therefore the average of the area of all these squares, here represented by the orange square.



Population versus Sample Variance

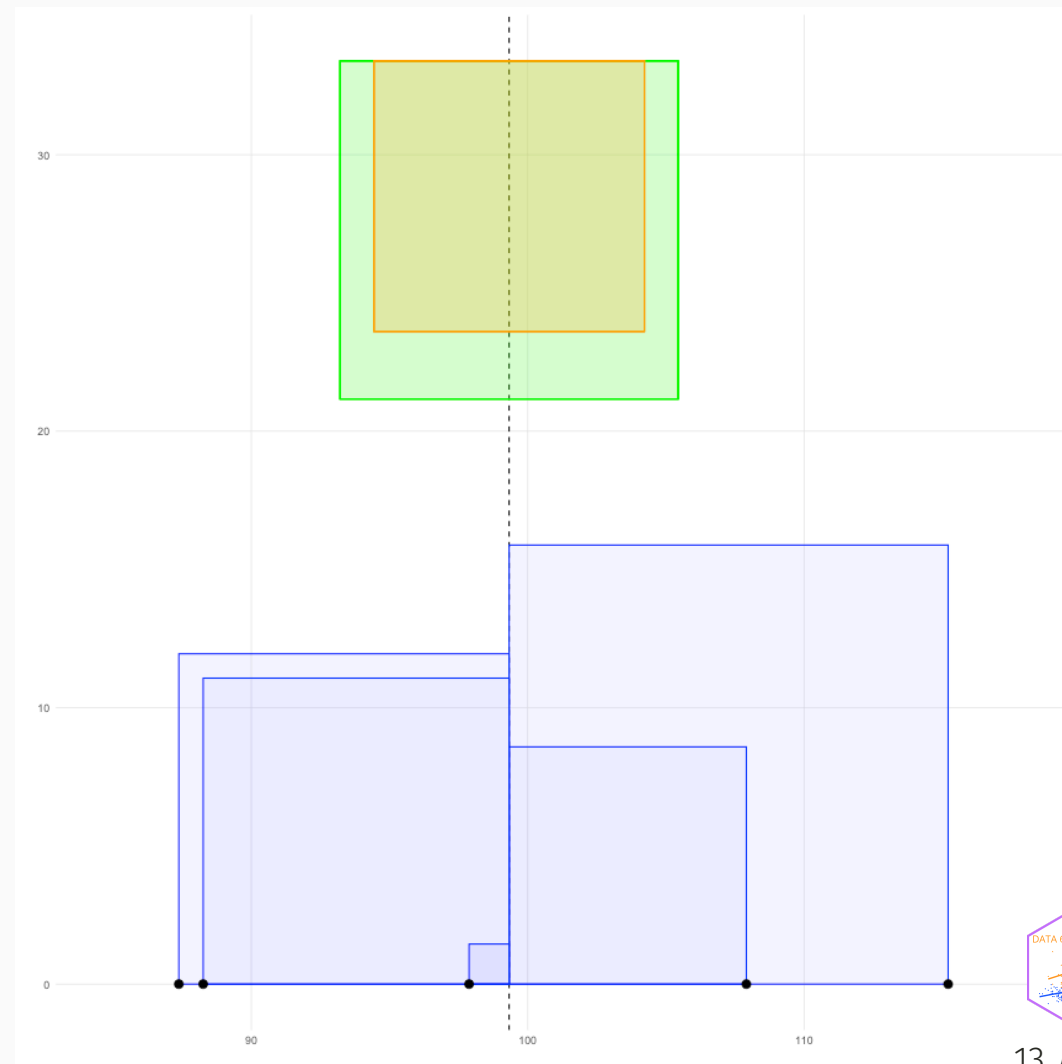
Typically we want the sample variance. The difference is we divide by $n - 1$ to calculate the sample variance. This results in a slightly larger area (variance) then if we divide by n .

Population Variance (yellow):

$$S^2 = \frac{\Sigma(x_i - \bar{x})^2}{N}$$

Sample Variance (green):

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1}$$



Robust Statistics

Consider the following data randomly selected from the normal distribution:

```
set.seed(41)
x <- rnorm(30, mean = 100, sd = 15)
mean(x); sd(x)
```

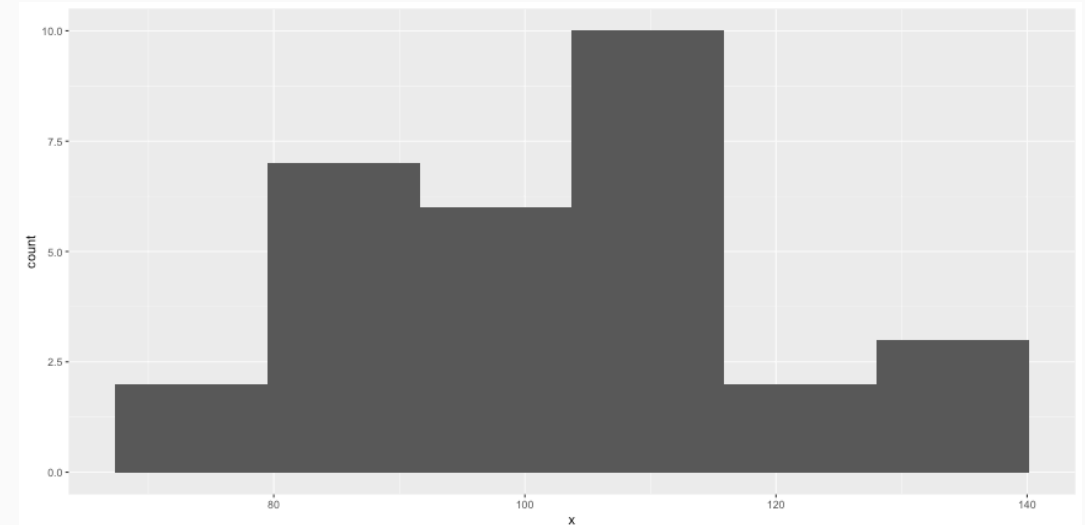
```
## [1] 103.1934
```

```
## [1] 16.8945
```

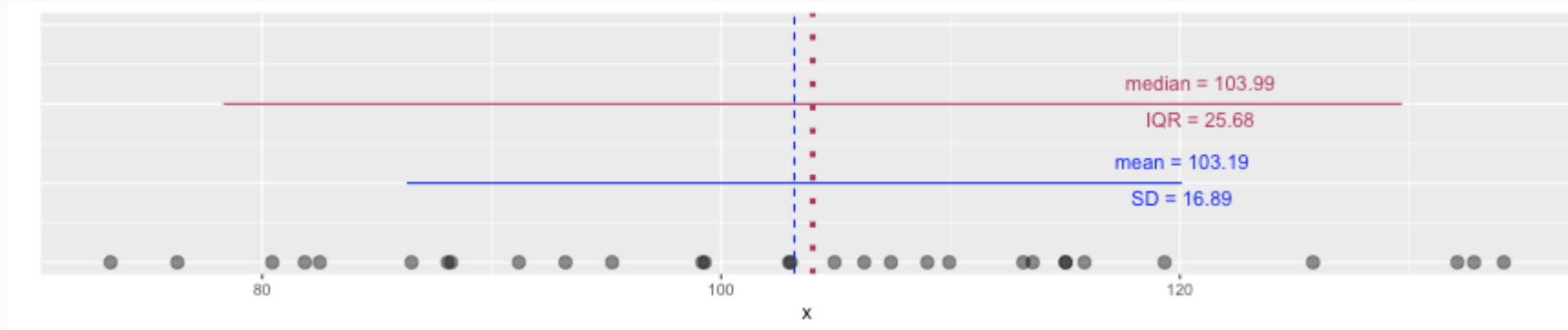
```
median(x); IQR(x)
```

```
## [1] 103.9947
```

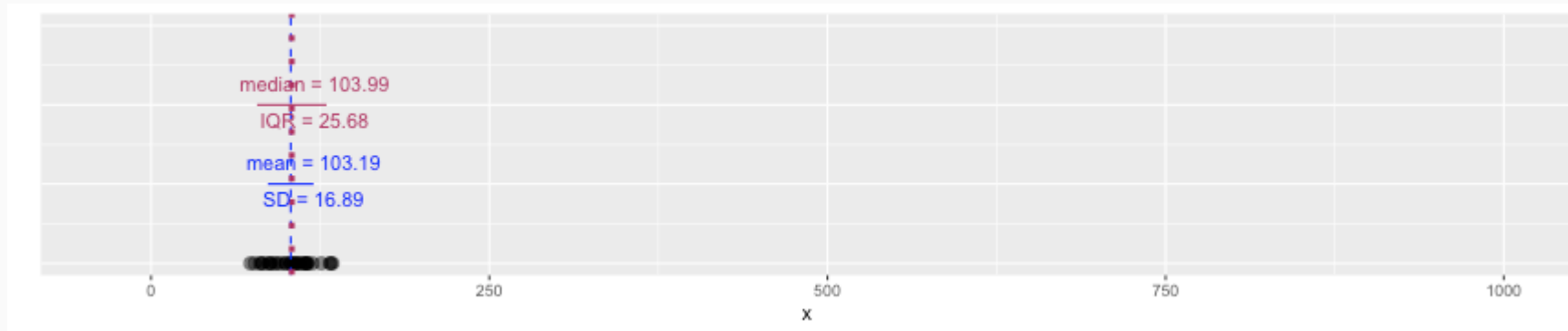
```
## [1] 25.68004
```



Robust Statistics

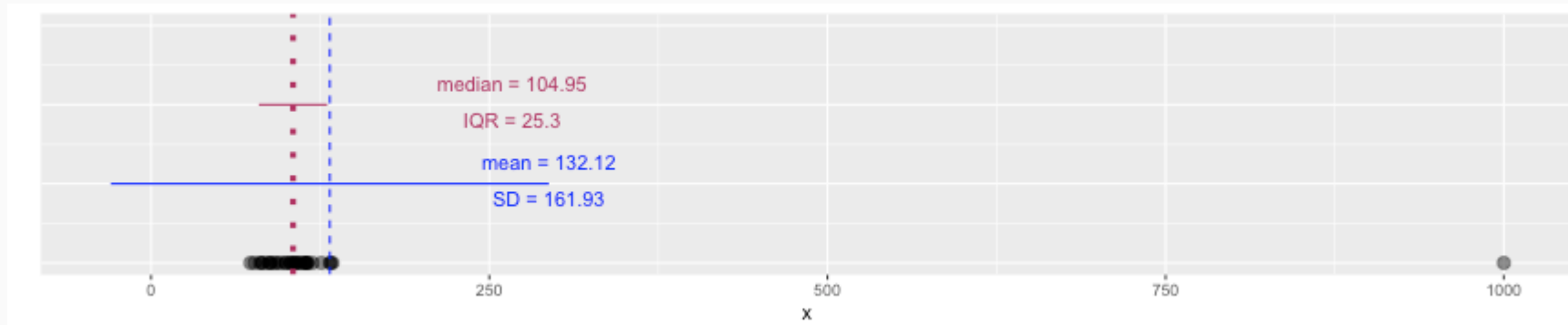


Robust Statistics



Let's add an extreme value:

```
x <- c(x, 1000)
```



Robust Statistics

Median and IQR are more robust to skewness and outliers than mean and SD. Therefore,

- for skewed distributions it is often more helpful to use median and IQR to describe the center and spread
- for symmetric distributions it is often more helpful to use the mean and SD to describe the center and spread

About legosets



To install the brickset package:

```
remotes::install_github('jbryer/brickset')
```

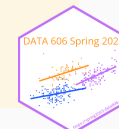
To load the load the legosets dataset.

```
data('legosets', package = 'brickset')
```

The legosets data has 21546 observations of 36 variables.

```
names(legosets)
```

```
## [1] "setID"          "number"          "numberVariant"
## [4] "name"           "year"            "theme"
## [7] "themeGroup"     "subtheme"        "category"
## [10] "released"       "pieces"          "minifigs"
## [13] "bricksetURL"    "rating"          "reviewCount"
## [16] "packagingType" "availability"     "agerange_min"
## [19] "thumbnailURL"   "imageUrl"        "US_retailPrice"
## [22] "US_dateFirstAvailable" "US_dateLastAvailable" "UK_retailPrice"
## [25] "UK_dateFirstAvailable" "UK_dateLastAvailable" "CA_retailPrice"
## [28] "CA_dateFirstAvailable" "CA_dateLastAvailable" "DE_retailPrice"
## [31] "DE_dateFirstAvailable" "DE_dateLastAvailable" "height"
## [34] "width"          "depth"           "weight"
```

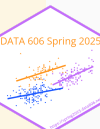


Structure (str)



```
str(legosets)
```

```
## 'data.frame':    21546 obs. of  36 variables:
## $ setID          : int  7693 7695 7697 7698 25534 7418 7419 6020 22704 7421 ...
## $ number         : chr  "1" "2" "3" "4" ...
## $ numberVariant   : int  8 8 6 4 6 1 1 1 3 4 ...
## $ name           : chr  "Small house set" "Medium house set" "Large house set" ...
## $ year           : int  1970 1970 1970 1970 1970 1970 1970 1970 1970 ...
## $ theme          : chr  "Minitalia" "Minitalia" "Minitalia" "Minitalia" ...
## $ themeGroup      : chr  "Vintage" "Vintage" "Vintage" "Vintage" ...
## $ subtheme       : chr  NA NA NA NA ...
## $ category       : chr  "Normal" "Normal" "Normal" "Normal" ...
## $ released       : logi  TRUE TRUE TRUE TRUE TRUE TRUE ...
## $ pieces         : int  67 109 158 233 NA 1 1 60 65 NA ...
## $ minifigs       : int  NA NA NA NA NA NA NA NA NA NA ...
## $ bricksetURL     : chr  "https://brickset.com/sets/1-8" "https://brickset.com/sets/2-8" "https://brickset.com/sets/3-6" "https://brickset.com/sets/4-4" ...
## $ rating         : num  0 0 0 0 0 0 0 0 0 ...
## $ reviewCount     : int  0 0 1 0 0 0 0 0 0 ...
## $ packagingType   : chr  "{Not specified}" "{Not specified}" "{Not specified}" "{Not specified}" ...
## $ availability    : chr  "{Not specified}" "{Not specified}" "{Not specified}" "{Not specified}" ...
## $ agerange_min    : int  NA NA NA NA NA NA NA NA NA NA ...
## $ thumbnailURL    : chr  "https://images.brickset.com/sets/small/1-8.jpg" "https://images.brickset.com/sets/small/2-8.jpg" "https://images.brickset.com/sets/small/3-6.jpg" "https://images.brickset.com/sets/small/4-4.jpg" ...
## $ imageURL       : chr  "https://images.brickset.com/sets/images/1-8.jpg" "https://images.brickset.com/sets/images/2-8.jpg" "https://images.brickset.com/sets/images/3-6.jpg" "https://images.brickset.com/sets/images/4-4.jpg" ...
## $ US_retailPrice  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ US_dateFirstAvailable: Date, format: NA NA ...
## $ US_dateLastAvailable : Date, format: NA NA ...
## $ UK_retailPrice  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ UK_dateFirstAvailable: Date, format: NA NA ...
## $ UK_dateLastAvailable : Date, format: NA NA ...
## $ CA_retailPrice  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ CA_dateFirstAvailable: Date, format: NA NA ...
## $ CA_dateLastAvailable : Date, format: NA NA ...
## $ DE_retailPrice  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ DE_dateFirstAvailable: Date, format: NA NA ...
## $ DE_dateLastAvailable : Date, format: NA NA ...
## $ height         : num  NA NA NA NA NA ...
## $ width          : num  NA NA NA NA NA ...
```



RStudio Environment tab can help



Environment	History	Connections	Git	Tutorial
R - Global Environment				
Data				
legosets 16355 obs. of 34 variables				
\$ setID	: int	7693 7695 7697 7698 25534 7418 7419 6020 22704 7421 ...		
\$ name	: chr	"Small house set" "Medium house set" "Medium house set" "L...		
\$ year	: int	1970 1970 1970 1970 1970 1970 1970 1970 1970 1970 ...		
\$ theme	: chr	"Minitalia" "Minitalia" "Minitalia" "Minitalia" ...		
\$ themeGroup	: chr	"Vintage" "Vintage" "Vintage" "Vintage" ...		
\$ subtheme	: chr	NA NA NA NA ...		
\$ category	: chr	"Normal" "Normal" "Normal" "Normal" ...		
\$ released	: logi	TRUE TRUE TRUE TRUE TRUE TRUE ...		
\$ pieces	: int	67 109 158 233 NA 1 1 60 65 NA ...		
\$ minifigs	: int	NA NA NA NA NA NA NA NA ...		
\$ bricksetURL	: chr	"https://brickset.com/sets/1-8" "https://brickset.com/sets...		
\$ rating	: num	0 0 0 0 0 0 0 0 0 ...		
\$ reviewCount	: int	0 0 1 0 0 0 0 1 0 0 ...		
\$ packagingType	: chr	"{Not specified}" "{Not specified}" "{Not specified}" "{No...		
\$ availability	: chr	"{Not specified}" "{Not specified}" "{Not specified}" "{No...		
\$ ageRange_min	: int	NA NA NA NA NA NA NA NA NA ...		
\$ US_retailPrice	: num	NA NA NA NA NA 1.99 NA NA 4.99 NA ...		
\$ US_dateFirstAvailable	: Date, format:	NA NA NA NA ...		
\$ US_dateLastAvailable	: Date, format:	NA NA NA NA ...		
\$ UK_retailPrice	: num	NA NA NA NA NA NA NA NA NA ...		
\$ UK_dateFirstAvailable	: Date, format:	NA NA NA NA ...		
\$ UK_dateLastAvailable	: Date, format:	NA NA NA NA ...		
\$ CA_retailPrice	: num	NA NA NA NA NA NA NA NA NA ...		
\$ CA_dateFirstAvailable	: Date, format:	NA NA NA NA ...		
\$ CA_dateLastAvailable	: Date, format:	NA NA NA NA ...		
\$ DE_retailPrice	: num	NA NA NA NA NA NA NA NA NA ...		
\$ DE_dateFirstAvailable	: Date, format:	NA NA NA NA ...		
\$ DE_dateLastAvailable	: Date, format:	NA NA NA NA ...		
\$ height	: num	NA NA NA NA NA ...		
\$ width	: num	NA NA NA NA NA ...		
\$ depth	: num	NA NA NA NA NA NA NA 5.08 NA ...		
\$ weight	: num	NA NA NA NA NA NA NA NA NA ...		
\$ thumbnailURL	: chr	"https://images.brickset.com/sets/small/1-8.jpg" "https://...		
\$ imageURL	: chr	"https://images.brickset.com/sets/images/1-8.jpg" "https://...		

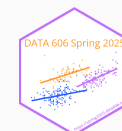


Table View

Show

10

 entries

Search:

	setID	name	year	theme	themeGroup	category	US_retailPrice	pieces	minifigs	rating
1	9788	Moose	2012	Promotional	Miscellaneous	Normal		27		0
2	28901	LEGO Minifigures - The LEGO Movie 2 Series {Random bag}	2019	Collectable Minifigures	Miscellaneous	Random				3.8
3	24155	Temple of Airjitzu	2015	Ninjago	Action/Adventure	Normal	199.99	2028	13	4.6
4	1324	Marie in Rainbow Skirt	1998	Scala	Girls	Normal		10	1	0
5	4184	Fire Engine	1995	Technic	Technical	Normal		424	2	3.6
6	28548	Kai Minifigure Alarm Clock	2018	Gear	Miscellaneous	Gear				0
7	31973	LEGO Meet the Minifigures	2022	Books	Miscellaneous	Book		8		0
8	4118	Universal Set with Flex System	1991	Technic	Technical	Normal		313		3.7
9	51241	Mario Kart - Mario & Standard Kart	2025	Super Mario	Licensed	Normal	169.99	1972		4.4
10	31879	Luigi Key Chain	2021	Gear	Miscellaneous	Gear				0

Showing 1 to 10 of 100 entries

Previous

1

2345...10Next

Data Wrangling Cheat Sheet



Data Transformation with dplyr : : CHEAT SHEET



dplyr functions work with pipes and expect **tidy data**. In tidy data:



Each **variable** is in its own **column**



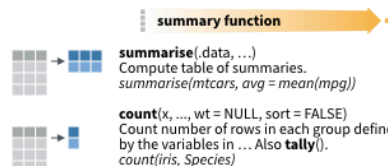
Each **observation**, or **case**, is in its own **row**



x %>% f(y) becomes **f(x, y)**

Summarise Cases

These apply **summary functions** to columns to create a new table of summary statistics. Summary functions take vectors as input and return one value (see back).

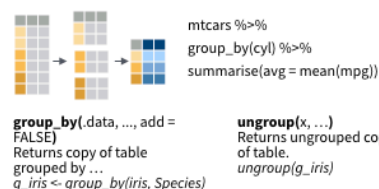


VARIATIONS

summarise_all() - Apply funs to every column.
summarise_at() - Apply funs to specific columns.
summarise_if() - Apply funs to all cols of one type.

Group Cases

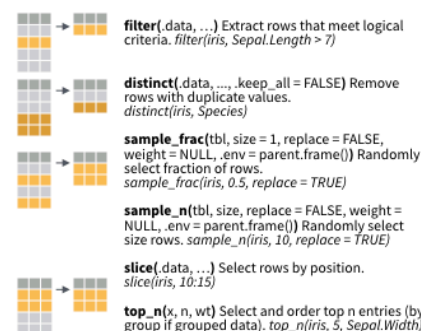
Use **group_by()** to create a "grouped" copy of a table. dplyr functions will manipulate each "group" separately and then combine the results.



Manipulate Cases

EXTRACT CASES

Row functions return a subset of rows as a new table.

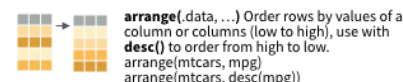


Logical and boolean operators to use with filter()

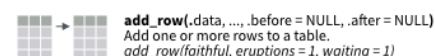
<	<=	is.na()	%in%		xor()
>	>=	!is.na()	!	&	

See **?base:logic** and **?Comparison** for help.

ARRANGE CASES



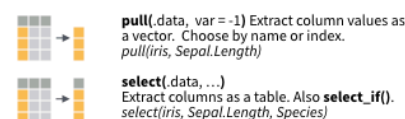
ADD CASES



Manipulate Variables

EXTRACT VARIABLES

Column functions return a set of columns as a new vector or table.

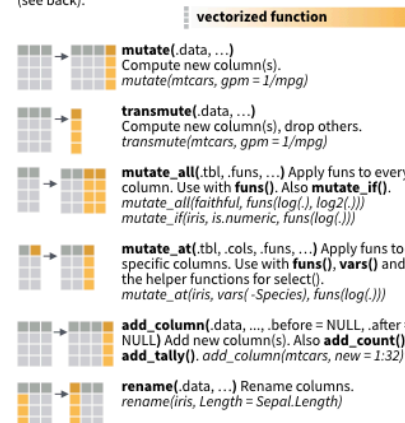


Use these helpers with **select()**, e.g. `select(iris, starts_with("Sepal"))`

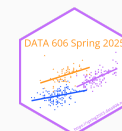
contains(match)	num_range(prefix, range)	z, e.g. mpg:cyl
ends_with(match)	one_of(...)	z, e.g. -Species
matches(match)	starts_with(match)	

MAKE NEW VARIABLES

These apply **vectorized functions** to columns. Vectorized funs take vectors as input and return vectors of the same length as output (see back).



RStudio® is a trademark of RStudio, Inc. • CC BY SA RStudio • info@rstudio.com • 844-448-1212 • rstudio.com • Learn more with `browseVignettes(package = c("dplyr", "tidbtle"))` • dplyr 0.7.0 • tidbtle 1.2.0 • Updated: 2017-03



R Syntax Comparison : : CHEAT SHEET

Dollar sign syntax

`goal(data$x, data$y)`

SUMMARY STATISTICS:

one continuous variable:
`mean(mtcars$mpg)`

one categorical variable:
`table(mtcars$cyl)`

two categorical variables:
`table(mtcars$cyl, mtcars$am)`

one continuous, one categorical:
`mean(mtcars$mpg[mtcars$cyl==4])`
`mean(mtcars$mpg[mtcars$cyl==6])`
`mean(mtcars$mpg[mtcars$cyl==8])`

PLOTTING:

one continuous variable:
`hist(mtcars$disp)`

`boxplot(mtcars$disp)`

one categorical variable:
`barplot(table(mtcars$cyl))`

two continuous variables:
`plot(mtcars$disp, mtcars$mpg)`

two categorical variables:
`mosaicplot(table(mtcars$am, mtcars$cyl))`

one continuous, one categorical:
`histogram(mtcars$disp[mtcars$cyl==4])`
`histogram(mtcars$disp[mtcars$cyl==6])`
`histogram(mtcars$disp[mtcars$cyl==8])`

`boxplot(mtcars$disp[mtcars$cyl==4])`
`boxplot(mtcars$disp[mtcars$cyl==6])`
`boxplot(mtcars$disp[mtcars$cyl==8])`

WRANGLING:

subsetting:
`mtcars[mtcars$mpg>30, ]`

making a new variable:
`mtcars$efficient[mtcars$mpg>30] <- TRUE`
`mtcars$efficient[mtcars$mpg<30] <- FALSE`

Formula syntax

`goal(y~x|z, data=data, group=w)`

SUMMARY STATISTICS:

one continuous variable:
`mosaic::mean(~mpg, data=mtcars)`

one categorical variable:
`mosaic::tally(~cyl, data=mtcars)`

two categorical variables:
`mosaic::tally(cyl~am, data=mtcars)`

one continuous, one categorical:
`mosaic::mean(mpg~cyl, data=mtcars)`

tilde

PLOTTING:

one continuous variable:
`lattice::histogram(~disp, data=mtcars)`

`lattice::bwplot(~disp, data=mtcars)`

one categorical variable:
`mosaic::bargraph(~cyl, data=mtcars)`

two continuous variables:
`lattice::xyplot(mpg~disp, data=mtcars)`

two categorical variables:
`mosaic::bargraph(~am, data=mtcars, group=cyl)`

one continuous, one categorical:
`lattice::histogram(~disp|cyl, data=mtcars)`
`lattice::bwplot(cyl~disp, data=mtcars)`

The variety of R syntaxes give you many ways to “say” the same thing

read across the cheatsheet to see how different syntaxes approach the same problem

Tidyverse syntax

`data %>% goal(x)`

SUMMARY STATISTICS:

one continuous variable:
`mtcars %>% dplyr::summarize(mean(mpg))`

one categorical variable:
`mtcars %>% dplyr::group_by(cyl) %>% dplyr::summarize(n())`

the pipe

two categorical variables:
`mtcars %>% dplyr::group_by(cyl, am) %>% dplyr::summarize(n())`

one continuous, one categorical:
`mtcars %>% dplyr::group_by(cyl) %>% dplyr::summarize(mean(mpg))`

PLOTTING:

one continuous variable:
`ggplot2::qplot(x=mpg, data=mtcars, geom = "histogram")`

`ggplot2::qplot(y=disp, x=1, data=mtcars, geom="boxplot")`

one categorical variable:
`ggplot2::qplot(x=cyl, data=mtcars, geom="bar")`

two continuous variables:
`ggplot2::qplot(x=disp, y=mpg, data=mtcars, geom="point")`

two categorical variables:
`ggplot2::qplot(x=factor(cyl), data=mtcars, geom="bar") + facet_grid(.~am)`

one continuous, one categorical:
`ggplot2::qplot(x=disp, data=mtcars, geom = "histogram") + facet_grid(.~cyl)`

`ggplot2::qplot(y=disp, x=factor(cyl), data=mtcars, geom="boxplot")`

WRANGLING:

subsetting:
`mtcars %>% dplyr::filter(mpg>30)`

making a new variable:
`mtcars <- mtcars %>% dplyr::mutate(efficient = if_else(mpg>30, TRUE, FALSE))`

Pipes %>% and |>



The pipe operator (%>%) introduced with the `magrittr` R package allows for the chaining of R operations. As of version 4.1, R now has a native pipe operator (|>). They take the output from the left-hand side and passes it as the first parameter to the function on the right-hand side.



You can do this in two steps:

```
tab_out <- table(legosets$category)
prop.table(tab_out)
```

Using the pipe (|>) operator we can chain these calls in a what is arguably a more readable format:

```
table(legosets$category) |> prop.table()
```

Or as nested function calls.

```
prop.table(table(legosets$category))
```

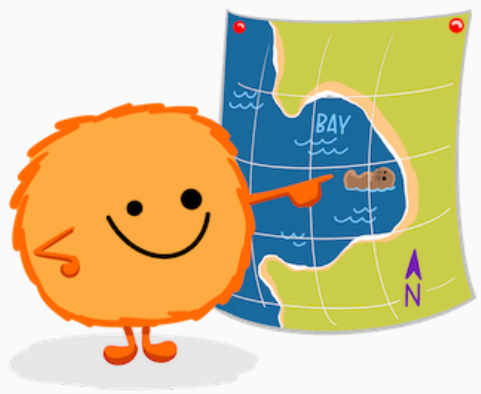
```
##
##      Book  Collection  Extended      Gear      Normal      Other
## 0.035087719 0.029889539 0.036248027 0.158869396 0.668894458 0.067205050
##      Random
## 0.003805811
```

dplyr::filter()

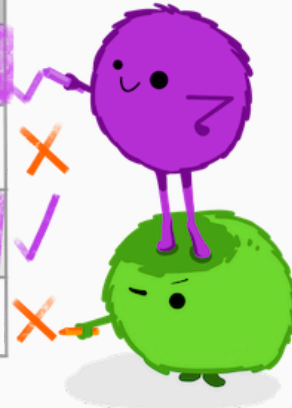
KEEP ROWS THAT
satisfy
your **CONDITIONS**

keep rows from... this data... ONLY IF... type MATCHES "otter" AND site MATCHES "bay"

```
filter(df, type == "otter" & site == "bay")
```



type	food	site
otter	urchin	bay
shark	seal	channel
otter	abalone	bay
otter	crab	wharf



@alison_horst

Logical Operators

- `!a` - TRUE if `a` is FALSE
- `a == b` - TRUE if `a` and `b` are equal
- `a != b` - TRUE if `a` and `b` are not equal
- `a > b` - TRUE if `a` is larger than `b`, but not equal
- `a >= b` - TRUE if `a` is larger or equal to `b`
- `a < b` - TRUE if `a` is smaller than `b`, but not equal
- `a <= b` - TRUE if `a` is smaller or equal to `b`
- `a %in% b` - TRUE if `a` is in `b` where `b` is a vector

```
which( letters %in% c('a','e','i','o','u') )
```

```
## [1] 1 5 9 15 21
```

- `a | b` - TRUE if `a` or `b` are TRUE
- `a & b` - TRUE if `a` and `b` are TRUE
- `isTRUE(a)` - TRUE if `a` is TRUE

dplyr

```
mylego <- legosets %>% filter(themeGroup == 'Educational' & year > 2015)
```

Base R

```
mylego <- legosets[legosets$themeGroup == 'Educational' & legosets$year > 2015,]
```

```
nrow(mylego)
```

```
## [1] 121
```

dplyr

```
mylego <- mylego %>% select(setID, pieces, theme, availability, US_retailPrice, minifigs)
```

Base R

```
mylego <- mylego[,c('setID', 'pieces', 'theme', 'availability', 'US_retailPrice', 'minifigs')]
```

```
head(mylego, n = 4)
```

##	setID	pieces	theme	availability	US_retailPrice	minifigs
## 1	26803	109	Education	{Not specified}	NA	6
## 2	26277	188	Education	Educational	94.95	NA
## 3	27742	160	Education	{Not specified}	NA	NA
## 4	26805	1000	Education	{Not specified}	NA	NA

Relocate



dplyr^{1.0.0}::relocate()
move COLUMNS around!

Default: move to FRONT
or move to
.before or .after
A SPECIFIED COLUMN!



@allison_horst



dplyr

```
mylego %>% relocate(where(is.numeric), .after = where(is.character)) %>% head(n = 3)
```

```
##           theme      availability setID pieces US_retailPrice minifigs
## 1 Education {Not specified} 26803    109           NA           6
## 2 Education      Educational 26277    188          94.95          NA
## 3 Education {Not specified} 27742    160           NA           NA
```

Base R

```
mylego2 <- mylego[,c('theme', 'availability', 'setID', 'pieces', 'US_retailPrice', 'minifigs')]
head(mylego2, n = 3)
```

```
##           theme      availability setID pieces US_retailPrice minifigs
## 1 Education {Not specified} 26803    109           NA           6
## 2 Education      Educational 26277    188          94.95          NA
## 3 Education {Not specified} 27742    160           NA           NA
```

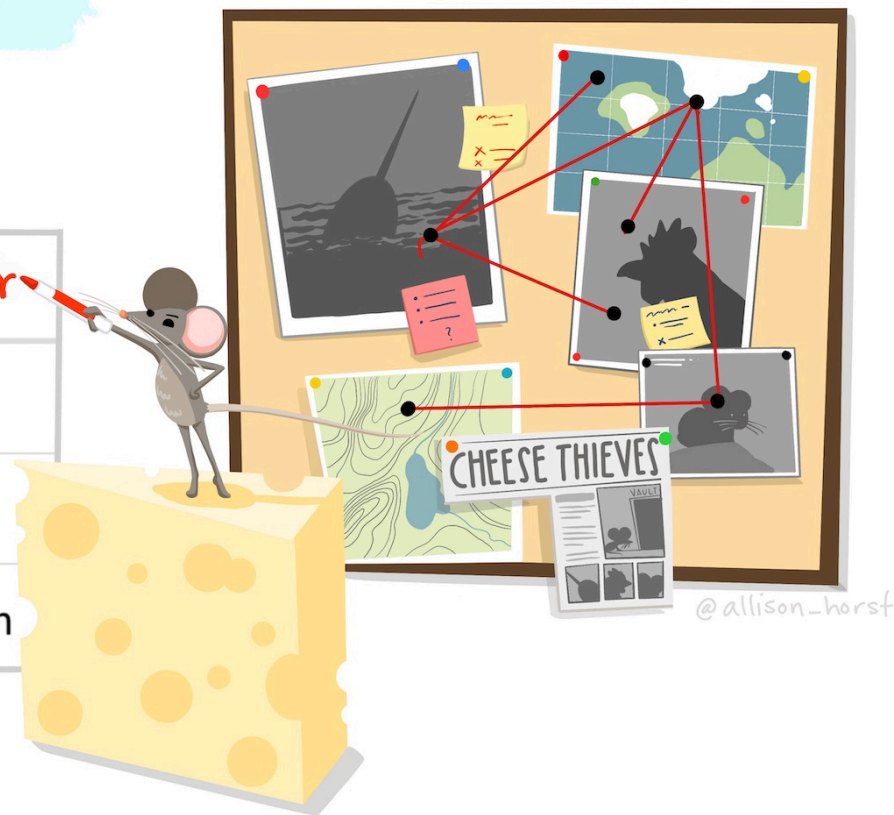
dplyr::rename()

RENAME COLUMNS*

```
df %>% rename(lair=site)
```

species nemesis	status	site lair
narwhal	unknown	ocean
chicken	active	coop
pika	active	mountain

*See `rename_with()` to rename using a function.



Rename



dplyr

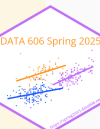
```
mylego %>% dplyr::rename(USD = US_retailPrice) %>% head(n = 3)
```

```
##   setID pieces   theme   availability   USD minifigs
## 1 26803   109 Education {Not specified}   NA         6
## 2 26277   188 Education   Educational 94.95        NA
## 3 27742   160 Education {Not specified}   NA        NA
```

Base R

```
names(mylego2)[5] <- 'USD'
head(mylego2, n = 3)
```

```
##           theme   availability setID pieces   USD minifigs
## 1 Education {Not specified} 26803   109   NA         6
## 2 Education   Educational 26277   188 94.95        NA
## 3 Education {Not specified} 27742   160   NA        NA
```



Mutate



dplyr

```
mylego %>% filter(!is.na(pieces) & !is.na(US_retailPrice)) %>%  
  mutate(Price_per_piece = US_retailPrice / pieces) %>% head(n = 3)
```

```
##   setID pieces    theme availability US_retailPrice minifigs Price_per_piece  
## 1 26277   188 Education Educational      94.95         NA      0.5050532  
## 2 25949   280 Education Educational     224.95         NA      0.8033929  
## 3 25954     1 Education Educational      14.95         NA     14.9500000
```

Base R

```
mylego2 <- mylego[!is.na(mylego$US_retailPrice) & !is.na(mylego$Price_per_piece),]  
mylego2$Price_per_piece <- mylego2$Price_per_piece / mylego2$US_retailPrice  
head(mylego2, n = 3)
```

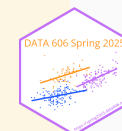
```
## [1] setID      pieces      theme      availability  
## [5] US_retailPrice minifigs    Price_per_piece  
## <0 rows> (or 0-length row.names)
```

Group By and Summarize



```
legosets %>% group_by(themeGroup) %>% summarize(mean_price = mean(US_retailPrice, na.rm = TRUE),
                                                sd_price = sd(US_retailPrice, na.rm = TRUE),
                                                median_price = median(US_retailPrice, na.rm = TRUE),
                                                n = n(),
                                                missing = sum(is.na(US_retailPrice)))
```

```
## # A tibble: 17 × 6
##   themeGroup      mean_price sd_price median_price      n missing
##   <chr>          <dbl>    <dbl>         <dbl> <int>  <int>
## 1 Action/Adventure    43.0    41.9          30.0  1620    845
## 2 Art and crafts     41.0    53.0          20.0   104     9
## 3 Basic              22.8    19.4          15.0   884   733
## 4 Constraction       16.4    12.4          13.0   503   285
## 5 Educational       185.    188.         138.   546   508
## 6 Girls              35.8    24.0          23.0   240   227
## 7 Historical         34.2    32.4          20.0   474   401
## 8 Junior             22.0    10.1          20.0   228   165
## 9 Licensed           55.5    72.9          35.0  3353  1245
## 10 Miscellaneous     23.4    33.5          15.0  7151  4501
## 11 Model making       86.2    99.4          50.0   919   422
## 12 Modern day        39.8    36.8          30.0  2669  1594
## 13 Pre-school        31.7    23.2          25.0  1613  1108
## 14 Racing            26.4    26.8          15.0   266   176
```



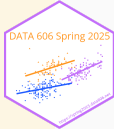
Describe and Describe By

```
library(psych)
describe(legosets$US_retailPrice)
```

```
##      vars      n mean      sd median trimmed      mad      min      max range skew kurtosis
## X1      1 8674 42.26 59.75  24.99   30.25 22.24 1.49 999.99 998.5 4.97    40.38
##
##      se
## X1 0.64
```

```
describeBy(legosets$US_retailPrice, group = legosets$availability, mat = TRUE, skew = FALSE)
```

```
##      item      group1 vars      n      mean      sd median      min      max range      se
## X11      1      {Not specified}      1 2004 27.76015 38.92310 19.99 1.49 789.99 788.5 0.8694780
## X12      2      Educational      1  12 217.03333 108.17617 232.45 14.95 399.95 385.0 31.2277699
## X13      3 Gift with Purchase at LEGO.com      1   0      NaN      NA      NA  Inf  -Inf  -Inf      NA
## X14      4      Insiders Reward      1   0      NaN      NA      NA  Inf  -Inf  -Inf      NA
## X15      5      LEGO exclusive      1 1268 65.12353 112.88247 14.99 1.99 999.99 998.0 3.1700555
## X16      6      LEGOLAND exclusive      1   2  4.99000  0.00000  4.99 4.99  4.99  0.0 0.00000000
## X17      7      Not sold      1   1 12.99000      NA 12.99 12.99 12.99  0.0      NA
## X18      8      Promotional      1   5  4.79000  0.83666  4.99 3.99  5.99  2.0 0.3741657
## X19      9      Promotional (Airline)      1   0      NaN      NA      NA  Inf  -Inf  -Inf      NA
## X110     10      Retail      1 5062 40.59401 40.96294 29.99 1.99 699.99 698.0 0.5757448
## X111     11      Retail - limited      1  319 63.40755 69.70365 39.99 2.49 449.99 447.5 3.9026552
```



Additional Resources

For data wrangling:

- dplyr website: <https://dplyr.tidyverse.org>
- R for Data Science book: <https://r4ds.had.co.nz/wrangle-intro.html>
- Wrangling penguins tutorial: <https://allisonhorst.shinyapps.io/dplyr-learnr/#section-welcome>
- Data transformation cheat sheet: <https://github.com/rstudio/cheatsheets/raw/master/data-transformation.pdf>

One Minute Paper

1. What was the most important thing you learned during this class?
2. What important question remains unanswered for you?



Scan the QR code or click here to
complete the One Minute Paper